Where Should a Robot Swarm See Its Neighbors? Communication-Range Transfer in a GNN Coverage Policy

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1 Application domain

Problem. In *coverage control*, N robots must spread over an area so that their sensors watch the regions that matter, described by an *importance density field* (IDF) ϕ that the robots do not know in advance [Cortés et al., 2004]. Each robot only sees a 64×64 m sensor footprint and can only exchange messages with robots within its *communication range* R; there is no central computer. The communication network is a graph $G_R = (V, E_R)$ with one node per robot and an edge (i, j) whenever $\|p_i - p_j\| \leq R$. This makes decentralized control a natural graph learning problem: every robot must decide what to send to its neighbors and how to combine what it receives [Tolstaya et al., 2020].

Dataset. We use the **CoverageControl** library [KumarRobotics, 2024] (GPL-3.0, `pip install coverage_control`), released with the LPAC paper [Agarwal et al., 2025]. It provides (i) a simulator of 32 robots in a 1024×1024 m world, with IDFs drawn as mixtures of 32 random Gaussians; (ii) the published LPAC checkpoint and the 11 evaluation environments it ships with; and (iii) scripts that generate imitation data by running a *clairvoyant* centroidal-Voronoi (CVT) controller that knows ϕ and all positions. One training sample is the state of all 32 robots at one time step: each robot's four 32×32 input maps, its position, the communication graph, and the clairvoyant's velocity as target. Following the paper we generate 100,000 samples per training set. We also generated and saved 40 new evaluation environments. Everything we use is public; our generated environments, data-generation configurations and code will be released in our repository[1] before the final submission.

Task and metric. The learned policy maps each robot's local observations and messages to a velocity, i.e. node-level regression on G_R, trained by imitation (mean squared error to the clairvoyant). We evaluate it in closed loop for 600 steps and report the paper's metric: the coverage cost $J = \int \phi(q) \min_i \|q - p_i\|^2 \, dq$ averaged over the episode, divided by the clairvoyant's cost on the same environment (lower is better; centralized CVT scores about 2.5). Our protocol varies the *test-time* range $R \in \{128, 256, 512, 768, 1024\}$ m, reports means over environments with paired differences, and uses three training seeds to set noise thresholds.

Why this dataset. LPAC is a recent published GNN policy whose authors report an unexplained anomaly: performance degrades when R grows (their Sec. VI-H). The simulator lets us control graph density exactly, the model is written in PyTorch Geometric [Fey and Lenssen, 2019], and a full training run takes 2 h 35 on one Stanford FarmShare GPU, so retraining several variants is affordable.

[1] `https://github.com/loicpoisson/lpac-communication-range` (made public before the final submission).

CS224W: Machine Learning with Graphs, Stanford University, Autumn 2026. Project proposal (type 1: real-world application).

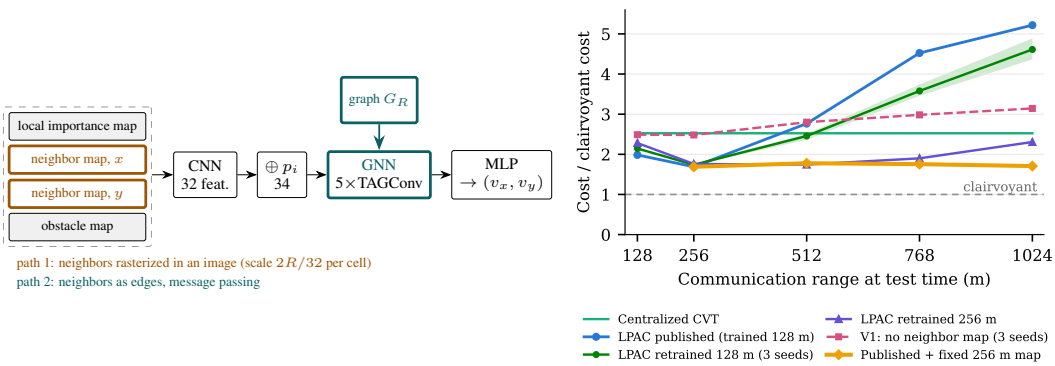


Figure 1: **Left:** LPAC on one robot. Neighbors enter twice: drawn in a 32×32 neighbor map read by the CNN (path 1) and as edges of G_R read by the GNN (path 2). **Right:** preliminary results, mean over the 11 released environments. Models trained at 128 m collapse when run with larger ranges; drawing the neighbor map at a fixed 256 m scale, without retraining, removes the collapse.

2 Graph ML techniques

Model: LPAC [Agarwal et al., 2025]. Each robot runs the same network (Fig. 1, left). A CNN encodes four 32×32 maps into 32 features, the robot position is appended, five TAGConv layers [Du et al., 2017] with $K = 3$ exchange information over G_R, and an MLP outputs the velocity. With node features X, adjacency A and degree matrix D,

$$X^{(\ell+1)} = \mathrm{ReLU}\Big(\textstyle\sum_{k=0}^{K} \big(D^{-1/2} A D^{-1/2}\big)^k X^{(\ell)} W_k^{(\ell)} \Big), \tag{1}$$

the GCN propagation [Kipf and Welling, 2017] applied up to 15 times in total. Two of the CNN's input channels form a *neighbor map*: robot j within range is written into cell $c_{ij} = \mathrm{round}\big((p_j - p_i)\frac{32}{2R} + 15.5\big)$ with value $(p_j - p_i)/R$, values summed per cell. GNNs suit this problem because message passing is exactly local communication: the policy is decentralized, permutation-equivariant, and independent of team size [Tolstaya et al., 2020].

Preliminary findings (motivate the project). (1) The published model (trained at 128 m) is the best controller at 128 and 256 m but its cost rises to 5.22 at 1024 m, worse than both CVT baselines, in all 11 environments; our three retrained models reproduce this (Fig. 1, right). (2) The GNN is not the culprit: at 1024 m, capping G_R to the 4 nearest neighbors leaves the cost at 5.11, while redrawing only the neighbor map at a fixed 256 m scale brings it to 1.71, the 256 m performance (paired difference $+0.02 \pm 0.16$); this holds on all our models and on 40 unseen environments. Effective rank [Roy and Vetterli, 2007] of the TAGConv outputs never collapses, ruling out over-smoothing. (3) Removing the neighbor map (V1) yields a flat but weaker policy (2.48 at 256 m vs 1.74).

Planned techniques. Our question is *where neighbor geometry should live*: in a rasterized image read by a CNN, or on the edges of the graph. We will compare, under identical data and training budgets: (a) inference-time interventions on each input path (graph capping, map rescaling); (b) LPAC trained on mixed communication ranges, a domain-randomization baseline [Tobin et al., 2017] (three seeds: no collapse, 1.9 at 1024 m, but 0.3 worse than LPAC at 256 m); (c) graph-native encoders that drop the neighbor map and carry the relative position $p_j - p_i$, normalized by a fixed length rather than by R, as edge attributes: GATv2 [Brody et al., 2022] (a first attempt underfits; we will tune it) and a PointNet-style edge convolution [Qi et al., 2017] placed before the TAGConv stack. Edge-based geometry is permutation-invariant and has no resolution, as in related swarm policies [Muthusamy et al., 2024, Owerko et al., 2025], and first results (three seeds) are promising but incomplete: with positions scaled by a fixed 256 m, the PointNet layer degrades little with range (2.01 at 256 m to 2.58 at 1024 m, against 1.74 to 4.61 for LPAC), but stays 0.3 behind LPAC at the training range and above the inference-time fix (1.8). We will close this gap with range-invariant encodings (unit direction with a saturating distance, geometry within a fixed radius) and by combining them with mixed-range training. Success: a variant within 0.05 of LPAC at 256 m that stays there up to 1024 m with no test-time fix (three seeds, released and new environments).

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